

# Bionics Engineering ; A. DeSilva

Mechanics

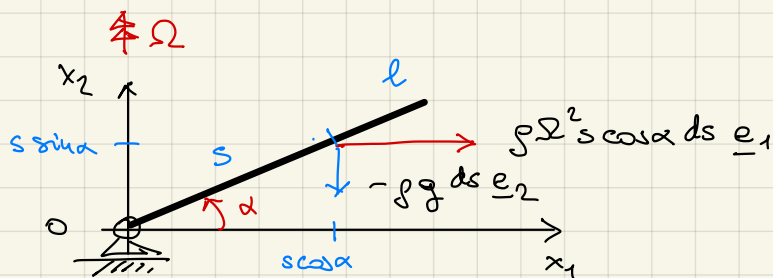
Bionics Eng

Minimal Potential Energy

and centrifugal force 

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## Revisit exercise on centrifugal forces in the light of PLV



$\underline{t} = (\cos \alpha, \sin \alpha)$  unit tangent vector

$$\underline{n} = \underline{e}_3 \wedge \underline{t} = (-\sin \alpha, \cos \alpha)$$

loads  $L_2 = \text{gravity}$ ,  $L_1 = \text{centrifugal force}$

Global equilibrium condition  $M_0 = 0$

$$\begin{aligned} M_0 &= \int_0^l (-p g s \cos \alpha - p \Omega^2 s^2 \cos \alpha \sin \alpha) ds \\ &= -p g \frac{l^2}{2} \cos \alpha - p \Omega^2 \frac{l^3}{3} \sin \alpha \cos \alpha = 0 \quad (*) \end{aligned}$$

### 1. PLV

Consider virtual displacement velocity field (rotation around  $\underline{e}_3$  at angular velocity  $\dot{\alpha}^*$ )

$$\dot{\underline{\xi}}^*(s) = \dot{\alpha}^* \underline{e}_3 \wedge s \underline{t} = \dot{\alpha}^* s \underline{n} = \dot{\alpha}^* s (-\sin \alpha, \cos \alpha)$$

$$\begin{aligned} P_{\text{ext}}[L_2, L_1](\dot{\underline{\xi}}^*) &= \int_0^l -p g \underline{e}_2 \cdot \underbrace{(\dot{\alpha}^* \underline{e}_3 \wedge s \underline{t})}_{\dot{\alpha}^* s (-\sin \alpha, \cos \alpha)} ds + \int_0^l p \Omega^2 s \cos \alpha \underline{e}_1 \cdot \underbrace{(\dot{\alpha}^* \underline{e}_3 \wedge s \underline{t})}_{\dot{\alpha}^* s (-\sin \alpha, \cos \alpha)} ds \\ &= \dot{\alpha}^* \int_0^l (-p g s \cos \alpha - p \Omega^2 s^2 \cos \alpha \sin \alpha) ds = \dot{\alpha}^* M_0 = 0 \quad \forall \dot{\alpha}^* \Rightarrow M_0 = 0 \end{aligned}$$

No assumptions on whether forces in  $L_2$ ,  $L_1$  are conservative has been made to derive equilibrium equation from PLV

## 2. Minimal potential energy

[Warning: this is more advanced! Assume  $\Omega$  is constant in time]

The loads  $L_a$  and  $L_b$  are conservative (positional forces coming from the gradient of a potential)

$$\underline{f}_a = -\rho g \underline{e}_2 = \nabla(-\rho g x_2) = \left( \frac{\partial}{\partial x_1}(-\rho g x_2), \frac{\partial}{\partial x_2}(-\rho g x_2) \right) = (0, -\rho g)$$

$$\underline{f}_b = \rho \Omega^2 x_1 \underline{e}_1 = \nabla \left( \rho \Omega^2 \frac{x_1^2}{2} \right) = \left( \frac{\partial}{\partial x_1} \left( \rho \Omega^2 \frac{x_1^2}{2} \right), \frac{\partial}{\partial x_2} \left( \rho \Omega^2 \frac{x_1^2}{2} \right) \right) = (\rho \Omega^2 x_1, 0)$$

$$\text{Define } U_a(\alpha) = \int_0^l -\rho g x_2(s) ds = \int_0^l -\rho g s \sin \alpha ds$$

$$U_b(\alpha) = \int_0^l \rho \Omega^2 \frac{x_1^2}{2} ds = \int_0^l \rho \Omega^2 s^2 \frac{\cos^2 \alpha}{2} ds$$

$$V(\alpha) = V_{\text{ext}}(\alpha) = - (U_a(\alpha) + U_b(\alpha)) \quad [\text{No elastic energy, } V = V_{\text{ext}}]$$

$$= \int_0^l \rho g s \sin \alpha ds - \int_0^l \rho \Omega^2 s^2 \frac{\cos^2 \alpha}{2} ds$$

$$\frac{d}{d\alpha} V(\alpha + h \underline{\alpha}^*) \Big|_{h=0} = \frac{d}{d\alpha} V(\alpha) \underline{\alpha}^*$$

$$\frac{d}{d\alpha} V(\alpha) = \int_0^l \rho g s \cos \alpha ds + \int_0^l \rho \Omega^2 s^2 \cos \alpha \sin \alpha ds$$

$$\underline{\alpha}^* \frac{d}{d\alpha} V(\alpha) = - \underline{\alpha}^* \left( \int_0^l (-\rho g s \cos \alpha - \rho \Omega^2 s^2 \sin \alpha \cos \alpha) \right) = -P_{\text{ext}}[L_a, L_b] \left( \frac{\underline{\alpha}^*}{2} \right) \text{ of 1.}$$

$$\Rightarrow \frac{d}{d\alpha} V(\alpha) = 0 \Leftrightarrow \underline{\alpha}^* \frac{d}{d\alpha} V(\alpha) = 0 \quad \forall \underline{\alpha}^* \Leftrightarrow P_{\text{ext}}[L_a, L_b] \left( \frac{\underline{\alpha}^*}{2} \right) = 0 \quad \forall \underline{\alpha}^*$$

We have used the fact that the forces are conservative to derive the equilibrium equations from the condition that a potential energy is stationary.  
The equilibrium equation is the same that we derived in 1, with PLV.

### 3. Special case $\Omega = 0$

In this case, only dead loads represent, and we recover the formulation of lecture 18

Assume as ref. config. the one with  $\alpha = 0$ .

$$\begin{aligned}\text{Then } V(\alpha) &= V_{\text{ext}}(\alpha) = -U_e(\alpha) = - \int_0^l (-\rho g s \sin \alpha) ds \\ &= - \int_0^l \underbrace{(-\rho g)}_{\text{force}} \underbrace{(s \sin \alpha - s \sin 0)}_{\text{vertical displacement in going from } \alpha=0 \text{ to } \alpha} ds \\ &= -W_{\text{ext}}(\alpha)\end{aligned}$$

Notice that

$$V(\alpha) = \int_0^l \rho g s \sin \alpha ds = \rho g \frac{l^2}{2} \sin \alpha$$

$$\frac{dV}{d\alpha} = \rho g \frac{l^2}{2} \cos \alpha = 0 \Rightarrow \alpha = \begin{cases} \pi/2 + 2k\pi \\ -\pi/2 + 2k\pi \end{cases} \quad \text{equilibrium configurations}$$

$$\frac{d^2V}{d\alpha^2} = -\rho g \frac{l^2}{2} \sin \alpha \begin{cases} < 0 \text{ at } \alpha = \pi/2 \text{ unstable equil.; here } V(\alpha) \text{ is max} \\ > 0 \text{ at } \alpha = -\pi/2 \text{ stable equil.; here } V(\alpha) \text{ is min.} \end{cases}$$

### Summary

- Can derive equilibrium eq. from PLV with no assumptions on the forces.
- If the forces are conservative, can derive equil. eq. from principle of stationary potential energy. This is nothing but a special case of PLV.
- When the conservative forces are in fact dead loads, their contribution to potential energy is  $-W_{\text{ext}}$
- Provided that  $\Omega$  is constant in time, centrifugal forces admit a potential and we can rederive the exercise with centrifugal forces using variational principles of minimal potential energy